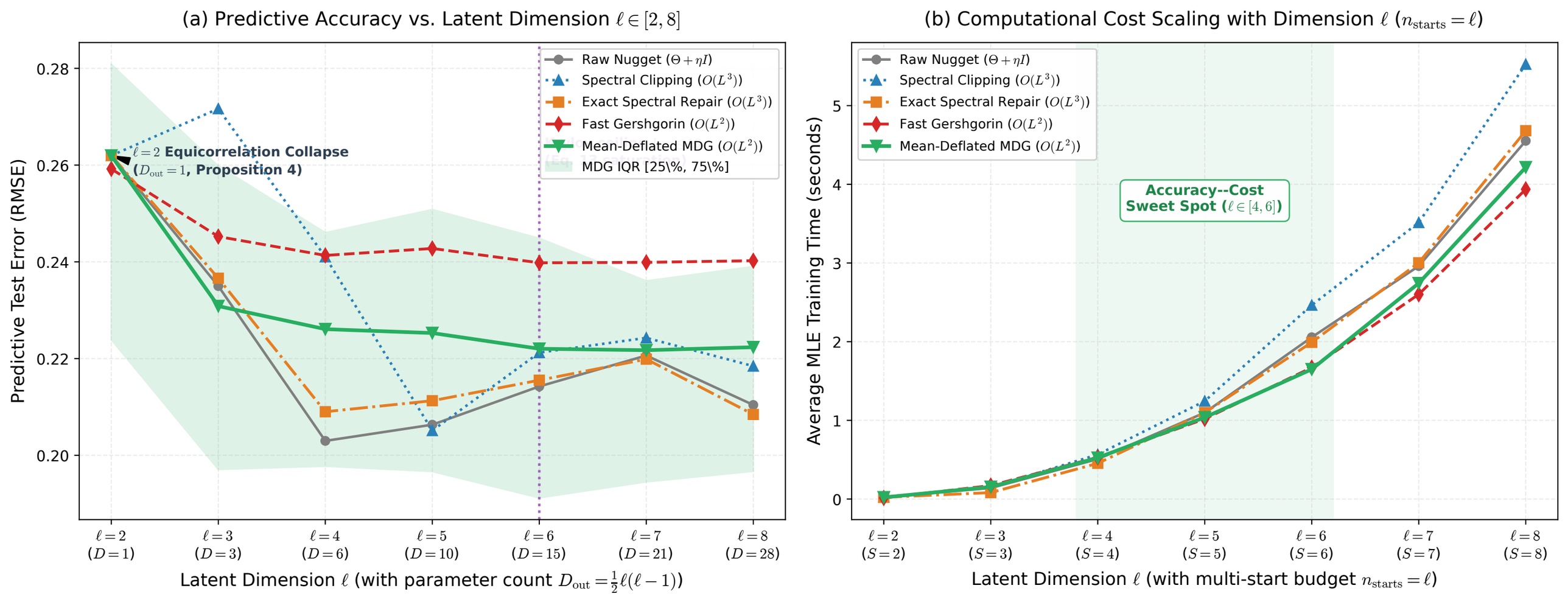
(a) Predictive Accuracy vs. Latent Dimension $\ell \in [2, 8]$
Predictive Test Error (RMSE)
0.28
0.26
0.24
0.22
0.20
Raw Nugget ($\Theta + \eta I$)
Spectral Clipping ($O(L^3)$)
Exact Spectral Repair ($O(L^3)$)
Fast Gershgorin ($O(L^2)$)
Mean-Deflated MDG ($O(L^2)$)
MDG IQR [25\%, 75\%]
$\ell = 2$ Equicorrelation Collapse
($D_{\text{out}} = 1$, Proposition 4)
$\ell = 2$ ($D = 1$)
$\ell = 3$ ($D = 3$)
$\ell = 4$ ($D = 6$)
$\ell = 5$ ($D = 10$)
$\ell = 6$ ($D = 15$)
$\ell = 7$ ($D = 21$)
$\ell = 8$ ($D = 28$)
Latent Dimension ℓ (with parameter count $D_{\text{out}} = \frac{1}{2}\ell(\ell - 1)$)
(b) Computational Cost Scaling with Dimension ℓ ($n_{\text{starts}} = \ell$)
Average MLE Training Time (seconds)
0
1
2
3
4
5
Raw Nugget ($\Theta + \eta I$)
Spectral Clipping ($O(L^3)$)
Exact Spectral Repair ($O(L^3)$)
Fast Gershgorin ($O(L^2)$)
Mean-Deflated MDG ($O(L^2)$)
Accuracy--Cost
Sweet Spot ($\ell \in [4, 6]$)
$\ell = 2$ ($S = 2$)
$\ell = 3$ ($S = 3$)
$\ell = 4$ ($S = 4$)
$\ell = 5$ ($S = 5$)
$\ell = 6$ ($S = 6$)
$\ell = 7$ ($S = 7$)
$\ell = 8$ ($S = 8$)
Latent Dimension ℓ (with multi-start budget $n_{\text{starts}} = \ell$)